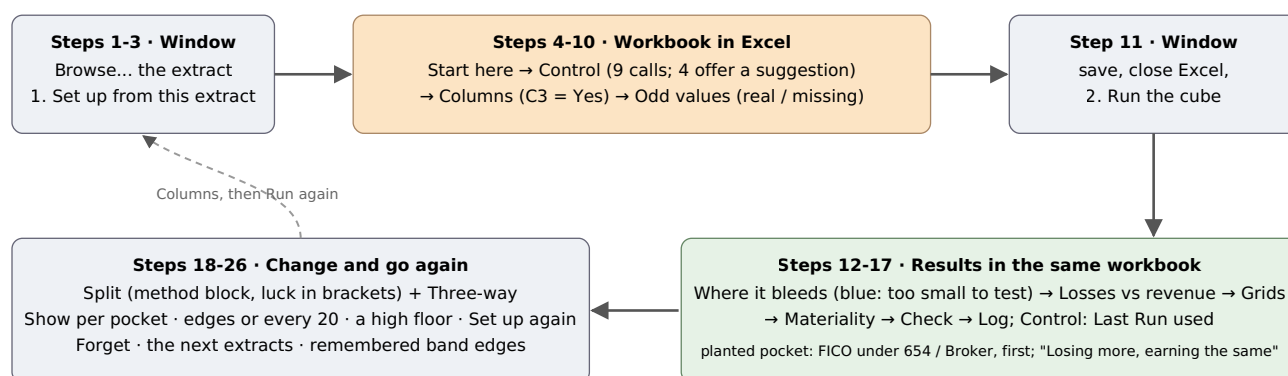


Origination analysis: the window and the workbook

Who this is for: the analyst preparing an origination analysis for a bank's line of business (LOB). No command line is needed. **What you finish with:** one workbook beside the loan extract. It holds your answers, a ranked list of the pockets that lose more than their share, losses set against revenue, and, if you ask for it, every pocket split by a third column and ranked again. **Walked:** 25 September 2026, on commit a5a8aa2, with a synthetic book of 8,000 loans. The book hides two things: a bad pocket (scores under 620 in the Broker channel), and borrowers carrying more revolving debt than usual for their score, who go bad 1.8 times as often. RANR has nothing planted in it. No real bank data was used.

What changed since the sixth walk:

- The suggested revenue option on Control is now **each pocket's own luck range**: revenue only counts as more or less when that pocket's own test says the gap isn't luck (step 13).
- On Losses vs revenue, a box marked "could be luck" isn't shaded and isn't counted with the findings. The count line gives those on their own. The RANR column is now **RANR reading**, read against the same line as the box.
- Pockets that are material but too small to test are **shaded blue** on Where it bleeds, and the window counts them (step 21).
- When nothing can be worked out from the book, the window and Control say the usual value was used, and the window says when no pocket had enough loans to test.
- Control's **Last Run used** column is on the printed page and stays through Set up (steps 9 and 22).
- Only band columns' edges are remembered, and Set up's count of columns to look at includes them (step 24).



Peach = where you answer (Excel). Grey = the window. Green = where you read the results. Nothing is typed at a command line.

The route: two buttons in the window, one workbook in Excel.

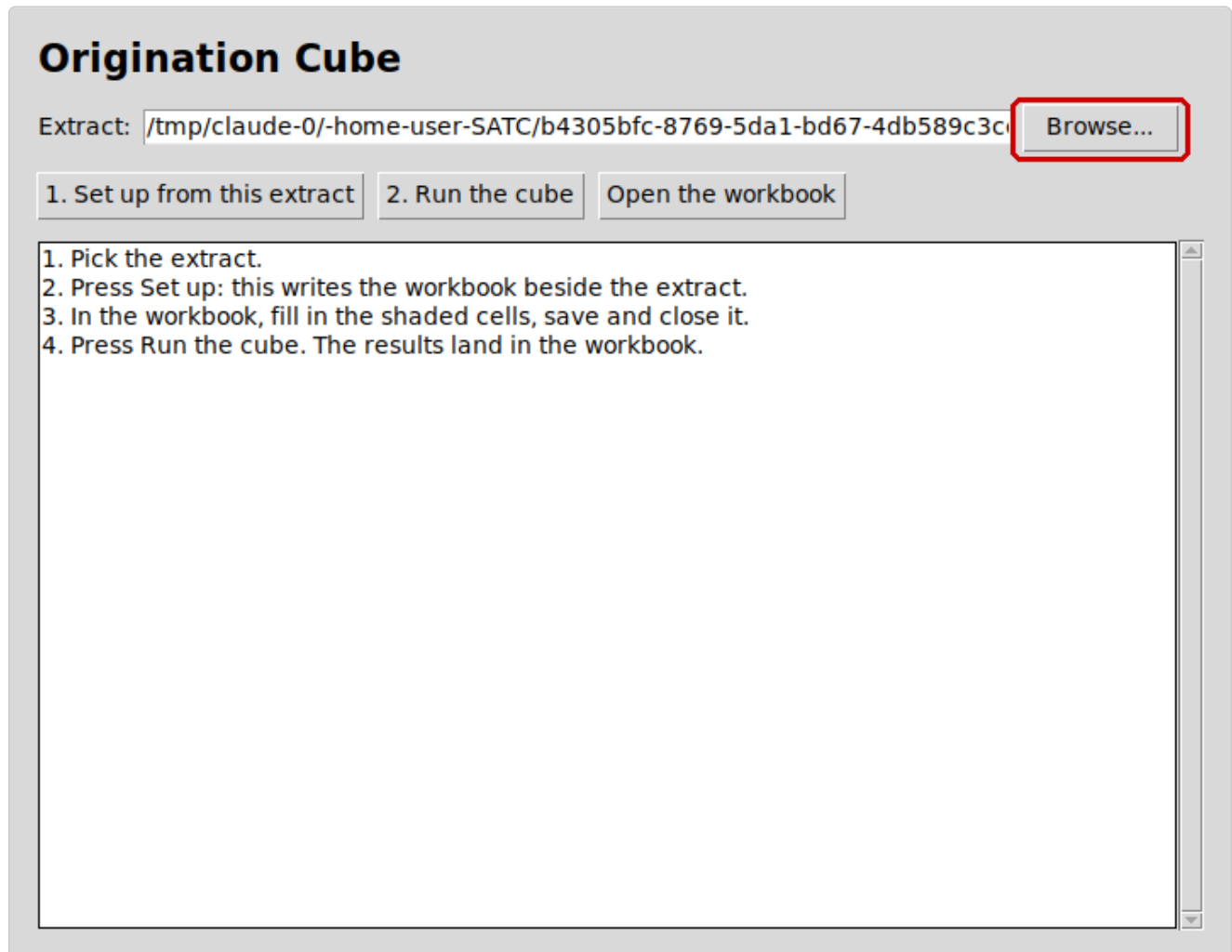
Before you start

- **Python 3.10 or later**, and **Install add-ons.bat** run once. You only need to do this the first time.
- **Excel.**

- **The loan extract** (.csv or .xlsx) in its own folder. It needs a loan number, the booked amount, a yes/no outcome, GCO dollars and RANR dollars.

Step 1: Open the window

Double-click **Origination Cube.pyw** in the origination-cube folder.



Right: a window titled Origination Cube, an Extract box, **Browse...**, three buttons, and four numbered lines in the box underneath.

Step 2: Pick the extract

Press **Browse...** and pick the loan file, not the workbook. On Windows this is the usual Open dialog.

Origination Cube

Extract:

1. Pick the extract.
2. Press Set up: this writes the workbook beside the extract.
3. In the workbook, fill in the shaded cells, save and close it.
4. Press Run the cube. The results land in the workbook.

Right: the file name shows at the right-hand end of the Extract box. The window remembers the last file picked, from any earlier session, so check it's this quarter's.

Step 3: Press 1. Set up from this extract

Origination Cube

Extract:

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns.

7 columns to look at first, on the Columns tab.

Next: fill in the shaded cells on Control, Columns and Odd values, save, close, and press Run.

Right: "Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns", then "7 columns to look at first", then what to fill in.

Check the loan count against what the bank sent. If it's wrong, you picked the wrong file.

Step 4: Open the workbook: Start here

Press **Open the workbook**. It opens on **Start here**.

Origination Cube

Set up from Consumer book Q3.csv: 8,000 loans, 9 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.

Where things stand

Calls still to make on Control	9
Things to look at first on Columns	7
Odd values found	2
Last run	not run yet

Right: four steps, and Where things stand: **9** calls to make on Control, 7 things to look at on Columns, 2 odd values, not run yet.

Step 5: Control: see what needs an answer

Click the **Control** tab.

Control center				
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.				
Shaded = still needs an answer				
Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				Needs an answer before we run. Pick one, or enter your own.
How old a loan must be to count				
How columns are recognised				
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				Needs an answer before we run. Pick one, or enter your own.
Fewest loans in a pocket before it is tested				Needs an answer before we run. Pick one, or enter your own.
Fewest loans with a loss before a loss rate is tested				Needs an answer before we run. Pick one, or enter your own.
Is it material				Needs an answer before we run. Pick one, or enter your own.
Smallest excess loss worth reporting				Needs an answer before we run. Pick one, or enter your own.
Does it underperform				Needs an answer before we run. Pick one from the list.
What a pocket is judged against		n/a		Needs an answer before we run. Pick one, or enter your own.
How much worse than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How much better than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How far revenue must move before it counts as more or less				Needs an answer before we run. Pick one, or enter your own.
Is it real				Needs an answer before we run. Pick one, or enter your own.
How sure a difference must be before it counts				Needs an answer before we run. Pick one, or enter your own.
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

Right: nine rows are shaded peach. They are the calls the tool won't make for you. Four offer a suggestion, marked (suggested) in their lists:

- **Fewest loans in a pocket before it is tested:** Enough for 5 expected losses.
- **How much worse** and **how much better:** What luck alone can move it: one line for every pocket, worked out from the book.
- **How far revenue must move** (ringed): also called What luck alone can move it, but it works differently. It means each pocket's own luck range, not one line. **The explanation beside it still describes the old single line** ("a pocket of typical size"). Go by this procedure, not that sentence (defect 2).

Nothing is picked for you.

Step 6: Columns: see what each column was taken to be

Click the **Columns** tab.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	0%	-9999, 713, 716
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	0%	Online, Broker, Branch
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	0%	7655.75, 27999.47, 39684.43
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	0%	1, 2, 4
REV_DEBT	Amount or number	Yes				suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)				
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes			suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.

Right: one row per column, with a meaning in What it is and the reason beside it. Seven rows are shaded under **Look first**. Checked every column? (C3) is empty. **Band edges** takes edges (620; 680) or a width (every 20).

Step 7: Odd values, and Learned

Click **Odd values**.

Odd values

Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.

Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	160		
RANR_AMT	negative values in a column that is mostly positive	2,612		

Right: FICO's -9999 (160 rows) and RANR's negatives (2,612 rows), each with an Answer cell.

On the first extract **Learned** says Nothing learned yet.

Learned

Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run, and the column waits on Columns for you to confirm it again.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
			Nothing learned yet.			

Step 8: If you press Run too early

Save, close and press **2. Run the cube** before answering, and nothing runs.

Origination Cube

Extract: -5da1-bd67-4db589c3cc31/scratchpad/w7/book/Consumer book Q3.csv

Browse...

1. Set up from this extract 2. Run the cube Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
- Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
- Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
- Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C21: "How far revenue must move before it counts as more or less" needs an answer. Pick one, or enter your own in column D.
- Control!C23: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.
- Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Right: "Couldn't run yet", then one line per thing to fix, naming the tab and cell. The same list goes to the **Log** tab.

Step 9: Fill in the shaded cells on Control

Click each shaded **Choose** cell and pick from its list, or type a number in **Or enter your own**. The walk took every suggestion:

Setting	Answer
How old a loan must be to count	Every loan

Setting	Answer
Fewest loans in a pocket before it is tested	Enough for 5 expected losses (suggested)
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse	What luck alone can move it (suggested)
How much better	What luck alone can move it (suggested)
How far revenue must move before it counts	What luck alone can move it (suggested)
How sure	95% sure

After the Run (step 11), Control's last column, **Last Run used**, shows what the Run actually used: **71** loans, **1.34** and **0.75** (worked out from this book), and revenue as **each pocket's own luck range**. It is inside the printed page now.

Control center					
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.					
Shaded = still needs an answer	Choose	Or enter your own	In use	What it means	Last Run used
What the cube looks at					
How old a loan must be to count	Every loan		Every loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.	Every loan
How columns are recognised					
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.	
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.	
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.	
Where the band edges fall	Each band holds about the same number of loans (recommended)	71/5	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.	
Is it enough loans to matter					
Fewest loans in a pocket before it is tested	Enough for 5 expected losses (suggested)		Enough for 5 expected losses (suggested)	Worked out from this book: the loans a pocket needs to expect 5 with the outcome at the book's rate, the textbook floor for testing a rate.	71 (worked out from this book)
Is it material					
Smallest excess loss worth reporting	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.	1% of the book's total losses
Does it underperform					
What a pocket is judged against	The rest of its band	71/5	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.	The rest of its band
How much worse than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the gap luck alone can make in a pocket of typical size. A smaller gap can't be told from noise.	1.34 (worked out from this book)
How much better than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	The same as above, the other way: one over the gap luck alone can make in a pocket of typical size.	0.75 (worked out from this book)
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the revenue gap luck alone can make in a pocket of typical size. Revenue inside that gap reads as about the same.	each pocket's own luck range
Is it real					
How sure a difference must be before it counts	95% sure		95% sure	The usual standard. About 1 pocket in 20 with no real difference will still pass.	95% sure
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.	80% of the time
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	71/5	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.	Hold down the share of lucky finds (Benjamini-Hochberg)
How much worse than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the gap luck alone can make in a pocket of typical size. A smaller gap can't be told from noise.	1.34 (worked out from this book)
How much better than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	The same as above, the other way: one over the gap luck alone can make in a pocket of 0.75 (worked out from this book) typical size.	0.75 (worked out from this book)
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the revenue gap luck alone can make in a pocket of typical size. Revenue inside that gap reads as about the same.	each pocket's own luck range
Is it real					
How sure a difference must be before it counts	95% sure		95% sure	The usual standard. About 1 pocket in 20 with no real difference will still pass.	95% sure

Four rows of Last Run used stay blank (the category limits, band count, band placement), though the Run used them.

Step 10: Confirm the columns and answer the odd values

On **Columns**, check each What it is, then set **Checked every column?** (C3) to **Yes**.

Columns

For any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done, confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan name contains 'bad'; 0 and 1 (7.1% are 1)	0%	7655.75, 27999.47, 39684.43 0, 2, 1
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes			suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

On **Odd values**, the walk said FICO's -9999 is **missing** and RANR's negatives are **real**.

Odd values

Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.

Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	160 missing		
RANR_AMT	negative values in a column that is mostly positive	2,612 real		

Step 11: Save, close, and press 2. Run the cube

Origination Cube

Extract: -5da1-bd67-4db589c3cc31/scratchpad/w7/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV DEBT 6,559 to under 9,837 / CHANNEL Broker.
Worked out from this book: fewest loans 71; worse at 1.34x; better at 0.75x.
Open Consumer book Q3 - Origination Cube.xlsx: start with where it bleeds.

Right:

- "Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree."
- one Worst for line per measure. For the three loss measures it's **FICO under 654 / CHANNEL Broker**, the planted pocket seen through five equal bands.
- "Worked out from this book: fewest loans 71; worse at 1.34x; better at 0.75x." (ringed).

The Worst for RANR line (REV_DEBT 6,559 to under 9,837 / Broker, RANR 0.74x) is noise: the test data plants nothing in RANR. One pocket in about a hundred passing by chance is what 95% sure allows.

Step 12: Read Where it bleeds

Where it bleeds																
Pockets losing more than their share (2020), warning loss, largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. Blue: material, but too few loans or losses to test, so worth a look by hand. "Luck alone" is how often a gap this big turns up with no real difference behind it, after the allowance for testing many pockets within each grid (see Checks).																
Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vo rest of book	Luck alone	Vo rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	90.7loans	yes	yes	4.14x under 0.01%	2.38x under 0.01% worse				1.47x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Broker	507	22.29%	7.13%	76.9loans	yes	yes	3.65x under 0.01%	2.01x under 0.01% worse				1.48x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	4	392	20.41%	7.13%	52.1loans	yes	yes	3.17x under 0.01%	1.54x 3.8% worse				1.55x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	4	424	18.63%	7.13%	48.8loans	yes	yes	2.87x under 0.01%	1.41x 8.2% worse, but could be luck				1.53x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	1	411	14.36%	7.13%	29.7loans	yes	yes	2.13x under 0.01%	0.98x 95.1% in line				1.53x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	3	350	15.14%	7.13%	28.1loans	yes	yes	2.24x under 0.01%	1.01x 98.1% in line				1.58x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	1	420	13.57%	7.13%	27.1loans	yes	yes	2.01x 0.02%	0.87x 67.5% in line				1.53x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	3	380	14.21%	7.13%	26.9loans	yes	yes	2.10x 0.01%	0.96x 93.2% in line				1.56x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	CHANNEL	Broker	537	11.73%	7.13%	24.7loans	yes	yes	1.73x 0.58%	1.58x 5.6% worse, but could be luck				1.46x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Branch	525	11.62%	7.13%	23.6loans	yes	yes	1.71x 0.14%	0.72x 6.3% better, but could be luck				1.47x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Online	568	10.56%	7.13%	19.5loans	yes	yes	1.54x 0.75%	0.63x 0.22% better				1.45x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET_CLASS	4	414	11.59%	7.13%	18.5loans	yes	yes	1.68x 2.4%	1.79x 6.4% worse, but could be luck				1.53x or more
Outcome, share of loans	FICO	under 654	CHANNEL	Online	531	10.36%	7.13%	17.2loans	yes	yes	1.50x 1.8%	0.59x 0.04% better				1.47x or more
Outcome, share of loans	FICO	654 to under 686	ASSET_CLASS	4	410	11.22%	7.13%	16.8loans	yes	yes	1.62x 1.2%	1.56x 12.5% worse, but could be luck				1.54x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	2	393	11.20%	7.13%	16.0loans	yes	yes	1.62x 1.5%	0.68x 9.0% better, but could be luck				1.55x or more
Outcome, share of loans	REV_DEBT	13,051 to under 17,187	ASSET_CLASS	4	411	10.96%	7.13%	15.7loans	yes	yes	1.58x 1.9%	1.51x 15.2% worse, but could be luck				1.53x or more
Outcome, share of loans	REV_DEBT	13,051 to under 17,187	CHANNEL	Broker	553	9.95%	7.13%	15.6loans	yes	yes	1.44x 2.8%	1.37x 22.3% worse, but could be luck				1.46x or more
Outcome, share of loans	FICO	under 654	CHANNEL	Branch	501	10.18%	7.13%	15.3loans	yes	yes	1.47x 2.9%	0.59x 0.04% better				1.48x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	2	385	10.91%	7.13%	14.6loans	yes	yes	1.57x 2.3%	0.69x 8.2% better, but could be luck				1.55x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET_CLASS	4	413	10.41%	7.13%	13.6loans	yes	yes	1.50x 8.6%	1.25x 39.8% in line				1.53x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	CHANNEL	Broker	531	9.23%	7.13%	11.2loans	yes	yes	1.32x 20.8%	1.30x 25.5% in line				1.47x or more
Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	ASSET_CLASS	4	412	9.71%	7.13%	10.6loans	yes	yes	1.39x 19.6%	1.80x 7.5% worse, but could be luck				1.53x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET_CLASS	3	427	9.37%	7.13%	9.6loans	yes	yes	1.34x 24.3%	1.08x 90.8% in line				1.52x or more
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Branch	503	8.95%	7.13%	9.2loans	yes	yes	1.28x 16.9%	1.13x 82.1% in line				1.48x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET_CLASS	1	404	9.16%	7.13%	8.2loans	yes	yes	1.31x 27.9%	1.04x 97.7% in line				1.54x or more
Outcome, share of loans	ORIG_BAL	49,152 and over	CHANNEL	Broker	520	8.46%	7.13%	6.9loans	yes	yes	1.20x 44.1%	1.90x 5.2% worse, but could be luck				1.47x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET_CLASS	3	382	8.90%	7.13%	6.8loans	yes	yes	1.26x 35.9%	1.19x 59.1% in line				1.56x or more
Outcome, share of loans	ORIG_BAL	under 15,449	CHANNEL	Broker	514	8.37%	7.13%	6.4loans	below the line	yes	1.19x 44.8%	1.42x 20.1% worse, but could be luck				1.47x or more
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Broker	531	8.18%	7.13%	5.2loans	below the line	yes	1.15x 42.6%	0.97x 87.3% in line				1.47x or more
Outcome, share of loans	ORIG_BAL	under 15,449	ASSET_CLASS	4	425	8.24%	7.13%	4.7loans	below the line	yes	1.17x 51.6%	1.34x 33.8% worse, but could be luck				1.53x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	CHANNEL	Branch	526	7.98%	7.13%	4.5loans	below the line	yes	1.13x 57.6%	1.03x 88.8% in line				1.47x or more
Outcome, share of loans	FICO	654 to under 686	ASSET_CLASS	3	377	8.22%	7.13%	4.1loans	below the line	yes	1.16x 51.8%	1.00x 98.1% in line				1.56x or more
Where it bleeds																
Pockets losing more than their share (2020), warning loss, largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. Blue: material, but too few loans or losses to test, so worth a look by hand. "Luck alone" is how often a gap this big turns up with no real difference behind it, after the allowance for testing many pockets within each grid (see Checks).																
Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vo rest of book	Luck alone	Vo rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	90.7loans	yes	yes	4.14x under 0.01%	2.38x under 0.01% worse				1.47x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Broker	507	22.29%	7.13%	76.9loans	yes	yes	3.65x under 0.01%	2.01x under 0.01% worse				1.48x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	4	392	20.41%	7.13%	52.1loans	yes	yes	3.17x under 0.01%	1.54x 3.8% worse				1.55x or more

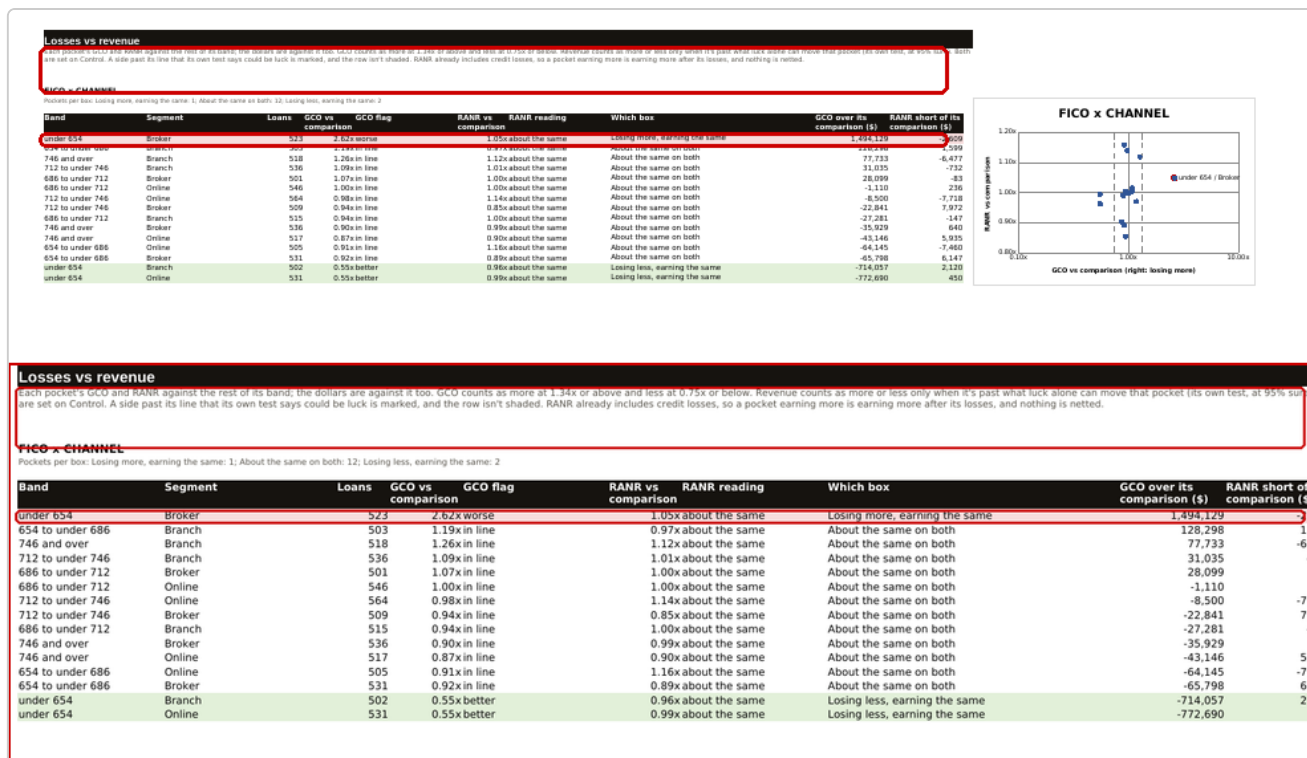
Right: the first row is FICO under 654 / Broker: 523 loans, 24.47% bad against a book of 7.13%, 90.7 loans of excess, 4.14x the rest of the book, 2.38x the rest of its band, **worse**.

How to read a row:

- **Red:** worse, and the test says it isn't luck. **Amber:** worse, but could be luck. **Blue:** material, but too few loans or losses to test; look at it by hand (step 21).
- **The flag** uses the comparison chosen in step 9 and the how much worse line (1.34x here).
- **Luck alone** is how often a gap this big turns up with no real difference, after the allowance for testing many pockets.
- **Smallest gap it could show** is what a pocket of that size can reliably find.

Step 13: Losses vs revenue

Every pocket at or above the fewest-loans line (71 here) goes in one of nine boxes. There's one block and one chart per grid.



Right:

- The subtitle gives the lines: GCO at **1.34x** and **0.75x**; "Revenue counts as more or less only when it's past what luck alone can move that pocket (its own test, at 95% sure)."
- FICO under 654 / Broker (ringed) reads **Losing more, earning the same**: GCO 2.62x the rest of its band, RANR 1.05x, about the same. Shaded red.
- RANR reading says about the same, earning more or earning less, against the same line as the box.

Count line and luck marks. A box can end in "(loss gap could be luck)". Those rows aren't shaded, and the count line lists them apart: "Marked could be luck, not counted above: 8". On this run 35 boxes carry a mark, all on the GCO side.

FICO x ASSET_CLASS

Pockets per box: Losing more, earning the same: 1; About the same on both: 11; Marked could be luck, not counted above: 8

Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
under 654	4	392	1.74x worse		1.03x about the same		Losing more, earning the same	643,768	-1,06
654 to under 686	4	410	1.68x worse, but could be luck	0.89x about the same	0.89x about the same		Losing more, earning the same (loss gap could be luck)	349,531	-96
712 to under 746	4	427	1.76x worse, but could be luck	1.07x about the same	1.07x about the same		Losing more, earning the same (loss gap could be luck)	182,349	-2,65
746 and over	2	399	1.37x worse, but could be luck	1.03x about the same	1.03x about the same		Losing more, earning the same (loss gap could be luck)	86,074	-1,53
686 to under 712	1	372	1.28x in line	0.94x about the same	0.94x about the same		About the same on both	78,056	2,16
654 to under 686	3	377	1.13x in line	1.19x about the same	1.19x about the same		About the same on both	68,431	-6,95
686 to under 712	4	407	1.18x in line	0.83x about the same	0.83x about the same		About the same on both	57,379	7,56
686 to under 712	3	388	1.14x in line	1.09x about the same	1.09x about the same		About the same on both	43,154	-3,27
712 to under 746	1	360	1.12x in line	1.00x about the same	1.00x about the same		About the same on both	28,137	16
746 and over	1	414	0.93x in line	1.13x about the same	1.13x about the same		About the same on both	-17,126	-5,58
746 and over	4	373	0.89x in line	0.98x about the same	0.98x about the same		About the same on both	-24,459	1,00
746 and over	3	385	0.83x in line	0.87x about the same	0.87x about the same		About the same on both	-43,059	5,95
under 654	3	350	0.95x in line	0.80x about the same	0.80x about the same		About the same on both	-46,186	8,05
712 to under 746	2	430	0.84x in line	0.84x about the same	0.84x about the same		About the same on both	-51,200	7,60
712 to under 746	2	302	0.48x better, but could be luck	1.13x about the same	1.13x about the same		Losing less, earning the same (loss gap could be luck)	-169,705	4,77

Where the box text wraps, it sits at the top of a taller row while the numbers sit at the bottom. Read the row by its numbers, not by the line the box text seems to be on (defect 4).

The chart plots the same pockets, with the GCO lines dashed. Under the suggested revenue option there are no revenue lines, since each pocket has its own. The three biggest Losing more pockets are named, including luck-marked ones:

of its
\$)

1,065

4,981

2,699

1,539

2,165

6,954

7,502

3,276

162

5,586

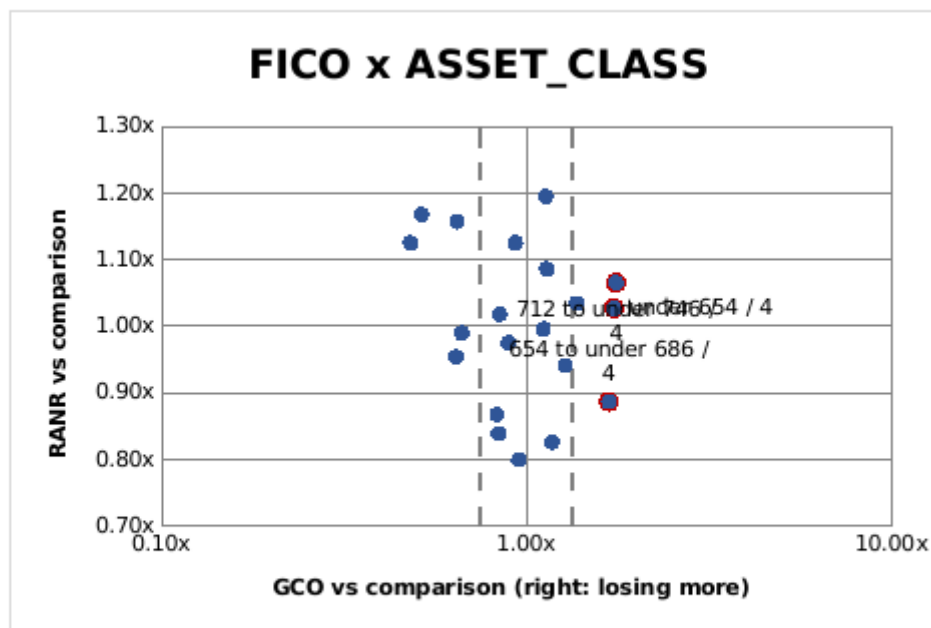
1,002

5,959

8,053

7,609

4,784



A pocket can read earning less on RANR noise when its own test says so. Here REV_DEBT 6,559 to under 9,837 / Broker, RANR 0.74x, is Losing the same, earning less, shaded red:

REV_DEBT x CHANNEL								
Pockets per box: Losing more, earning the same: 1; Losing the same, earning less: 1; About the same on both: 8. Marked could be luck, not counted above: 5								
Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)
17,187 and over	Broker	507	1.84x worse		1.09x about the same		Losing more, earning the same	865,403
13,051 to under 17,187	Broker	553	1.65x worse, but could be luck		0.94x about the same		Losing more, earning the same (loss gap could be luck)	-4,13
9,837 to under 13,051	Broker	532	1.32x in line		1.11x about the same		About the same on both	431,405
9,837 to under 13,051	Branch	540	1.21x in line		1.05x about the same		About the same on both	148,541
under 6,559	Broker	541	1.16x in line		0.97x about the same		About the same on both	101,121
under 6,559 to under 9,837	Broker	522	1.12x in line		0.74x earning less		Losing the same, earning less	48,582
under 6,559	Branch	522	1.07x in line		0.89x about the same		About the same on both	40,214
6,559 to under 9,837	Online	564	0.98x in line		1.06x about the same		About the same on both	20,381
6,559 to under 9,837	Branch	513	0.91x in line		1.25x about the same		About the same on both	-8,320
under 6,559	Online	537	0.80x in line		1.14x about the same		About the same on both	-32,100
13,051 to under 17,187	Online	522	0.87x in line		1.03x about the same		About the same on both	-68,849
9,837 to under 13,051	Online	527	0.58x better, but could be luck		0.86x about the same		Losing less, earning the same (loss gap could be luck)	-103,586
13,051 to under 17,187	Branch	525	0.64x better, but could be luck		1.04x about the same		Losing less, earning the same (loss gap could be luck)	-250,443
17,187 and over	Branch	525	0.75x better, but could be luck		0.87x about the same		Losing less, earning the same (loss gap could be luck)	-311,599
17,187 and over	Online	568	0.68x better, but could be luck		1.05x about the same		Losing less, earning the same (loss gap could be luck)	-373,208
								-529,252

Other revenue lines. Change How far revenue must move on Control, save, close and Run. The fixed options are plain lines, the same for every pocket, with the luck mark:

Revenue option	Lines	Boxes marked luck	Planted pocket, default bands	under 620 / Broker, 620; 680; 740	600 to under 620 / Broker, every 20
Each pocket's own luck range (suggested)	its own test	35	Losing more, earning the same (red)	Losing more, earning the same (red)	Losing more, earning the same (red)
5 percent either way	1.05x / 0.95x	73	the same (red)	Losing more, earning more (revenue gap could be luck): not shaded, not counted	the same: not shaded, not counted
		58			

Revenue option	Lines	Boxes marked luck	Planted pocket, default bands	under 620 / Broker, 620; 680; 740	600 to under 620 / Broker, every 20
10 percent either way	1.10x / 0.90x		the same (red)	not shaded, not counted	not shaded, not counted
The same lines as for losses	1.33x / 0.75x	35	the same (red)	Losing more, earning the same (red)	Losing more, earning the same (red)
Your own: 0.15	1.15x / 0.85x	43	the same (red)	not shaded, not counted	not shaded, not counted

Under 5%, 10% or your own line, check every "(revenue gap could be luck)" row's GCO flag. If it says worse, the pocket is a real loss finding even though the row isn't shaded or counted. With every 20 and 5%, all three real worse rows on the tab are unshaded (defect 1).

With 5% either way, default bands:

Losses vs revenue									
each pocket's GCO and RANR against the rest of its band; the dollars are against it too. GCO counts as more at 1.34x or above and less at 0.95x or below. Revenue counts as more at 1.05x or above and less at 0.95x or below either way. Both are set on Control. A 1.0x line that its own test says could be luck is marked, and the row isn't shaded. RANR already includes credit losses, so a pocket earning more is earning more after its losses, and nothing is netted.									
FICO x CHANNEL									
pocket's pay box: Losing more, earning the same: 1. About the same on both: 6. Losing less, earning the same: 2. Marked could be luck, not counted above 6									
Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
under 654	Broker	523	2.62x worse		1.05x about the same	Losing more, earning the same		1,494,129	-2,609
654 to under 686	Branch	503	1.19x in line		0.97x about the same	About the same on both		128,298	1,599
746 and over	Branch	518	1.26x in line		1.12x earning more (could be luck)	Losing the same, earning more (revenue gap could be		77,733	-6,477
712 to under 746	Branch	536	1.09x in line		1.01x about the same	About the same on both		31,035	-732
686 to under 712	Broker	501	1.07x in line		1.00x about the same	About the same on both		28,099	-83
686 to under 712	Online	546	1.00x in line		1.00x about the same	About the same on both		-1,110	236
712 to under 746	Online	564	0.98x in line		1.14x earning more (could be luck)	Losing the same, earning more (revenue gap could be		-8,500	-7,732
712 to under 746	Broker	509	0.94x in line		0.85x earning less (could be luck)	Losing the same, earning less (revenue gap could be		-22,841	7,972
686 to under 712	Branch	515	0.94x in line		1.00x about the same	About the same on both		-27,281	-147
746 and over	Branch	536	0.90x in line		0.99x about the same	Losing the same, earning less (revenue gap could be		-35,929	640
746 and over	Online	517	0.87x in line		0.90x earning less (could be luck)	Losing the same, earning more (revenue gap could be		-43,146	5,935
654 to under 686	Online	505	0.91x in line		1.16x earning more (could be luck)	Losing the same, earning less (revenue gap could be		-64,145	-7,460
654 to under 686	Broker	531	0.92x in line		0.89x earning less (could be luck)	Losing the same, earning less (revenue gap could be		-65,798	6,147
under 654	Branch	502	0.55x better		0.96x about the same	Losing less, earning the same		-714,037	2,120
under 654	Online	531	0.55x better		0.99x about the same	Losing less, earning the same		-772,690	450

FICO x CHANNEL

Losses vs revenue									
each pocket's GCO and RANR against the rest of its band; the dollars are against it too. GCO counts as more at 1.34x or above and less at 0.95x or below. Revenue counts as more at 1.05x or above and less at 0.95x or below either way. Both are set on Control. A 1.0x line that its own test says could be luck is marked, and the row isn't shaded. RANR already includes credit losses, so a pocket earning more is earning more after its losses, and nothing is netted.									
FICO x CHANNEL									
pocket's pay box: Losing more, earning the same: 1. About the same on both: 6. Losing less, earning the same: 2. Marked could be luck, not counted above 6									
Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
under 654	Broker	523	2.62x worse		1.05x about the same	Losing more, earning the same		1,494,129	-2,609
654 to under 686	Branch	503	1.19x in line		0.97x about the same	About the same on both		128,298	1,599
746 and over	Branch	518	1.26x in line		1.12x earning more (could be luck)	Losing the same, earning more (revenue gap could be		77,733	-6,477
712 to under 746	Branch	536	1.09x in line		1.01x about the same	About the same on both		31,035	-732
686 to under 712	Broker	501	1.07x in line		1.00x about the same	About the same on both		28,099	-83
686 to under 712	Online	546	1.00x in line		1.00x about the same	About the same on both		-1,110	236
712 to under 746	Online	564	0.98x in line		1.14x earning more (could be luck)	Losing the same, earning more (revenue gap could be		-8,500	-7,732
712 to under 746	Broker	509	0.94x in line		0.85x earning less (could be luck)	Losing the same, earning less (revenue gap could be		-22,841	7,972
686 to under 712	Branch	515	0.94x in line		1.00x about the same	About the same on both		-27,281	-147
746 and over	Branch	536	0.90x in line		0.99x about the same	Losing the same, earning less (revenue gap could be		-35,929	640
746 and over	Online	517	0.87x in line		0.90x earning less (could be luck)	Losing the same, earning less (revenue gap could be		-43,146	5,935
654 to under 686	Online	505	0.91x in line		1.16x earning more (could be luck)	Losing the same, earning more (revenue gap could be		-64,145	-7,460
654 to under 686	Broker	531	0.92x in line		0.89x earning less (could be luck)	Losing the same, earning less (revenue gap could be		-65,798	6,147

Your own value is a share up to 0.9. 95 is refused with "for 15%, type 0.15"; 0.95 with "The most it takes is 0.9."

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Control!D21: "How far revenue must move before it counts as more or less" needs a share, such as 0.1 for 10%, from 0.01 to 0.9; got 95. Type a share between 0.01 and 0.9: for 15%, type 0.15.

Origination Cube

Extract: db589c3cc31/scratchpad/w7/wrong/rev-own095/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!D21: "How far revenue must move before it counts as more or less" needs a share, such as 0.1 for 10%, from 0.01 to 0.9; got 0.95. The most it takes is 0.9.

Step 14: Grids

Grids shows each band crossed with each segment, one block per measure. Red is worse, green is better; for RANR low is red.

Grids					Grids					Grids				
FICO x CHANNEL: Outcome, share of loans					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	10.18%	24.47%	10.36%	11.00%	under 654	1.43x	1.43x	1.45x	2.11x	under 654	0.59x	2.38x	2.11x	
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x	654 to under 686	1.13x	0.97x	0.92x	
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x	686 to under 712	1.16x	1.08x	1.08x	
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x	712 to under 746	1.15x	0.79x	1.07x	
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x	746 and over	1.20x	0.88x	0.90x	
(blank)					(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)					(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x	All				
FICO x CHANNEL: Outcome, share of booked dollars					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	8.43%	21.52%	9.80%	14.92%	under 654	1.43x	1.35x	1.47x	2.11x	under 654	0.60x	2.41x	2.11x	
654 to under 686	8.43%	7.61%	7.42%	7.86%	654 to under 686	1.26x	1.14x	1.07x	1.14x	654 to under 686	1.14x	0.96x	0.96x	
686 to under 712	4.62%	4.63%	4.31%	4.50%	686 to under 712	0.68x	0.67x	0.63x	0.68x	686 to under 712	1.05x	1.04x	0.96x	
712 to under 746	4.12%	3.55%	3.78%	3.62%	712 to under 746	0.49x	0.52x	0.54x	0.51x	712 to under 746	1.12x	0.80x	0.86x	
746 and over	4.48%	3.37%	3.46%	3.68%	746 and over	0.59x	0.49x	0.52x	0.53x	746 and over	1.18x	0.88x	0.97x	
(blank)					(blank)					(blank)				
(marked missing)	1.68%	10.22%	2.93%	4.82%	(marked missing)					(marked missing)				
All	6.23%	8.82%	5.69%	6.93%	All	0.90x	1.28x	0.82x	1.00x	All				
FICO x CHANNEL: GCD per booked dollar					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	5.48%	14.11%	5.43%	8.15%	under 654	1.43x	1.47x	1.45x	2.11x	under 654	0.60x	2.41x	2.11x	
654 to under 686	5.83%	4.23%	4.20%	4.41%	654 to under 686	1.26x	1.14x	1.08x	1.14x	654 to under 686	1.14x	0.96x	0.96x	
686 to under 712	2.42%	2.44%	2.52%	2.53%	686 to under 712	0.68x	0.68x	0.68x	0.68x	686 to under 712	0.84x	1.07x	1.05x	
712 to under 746	2.33%	2.08%	2.14%	2.17%	712 to under 746	0.49x	0.54x	0.56x	0.54x	712 to under 746	1.08x	0.84x	0.86x	
746 and over	2.34%	1.88%	1.77%	1.84%	746 and over	0.59x	0.47x	0.46x	0.50x	746 and over	1.26x	0.86x	0.89x	
(blank)					(blank)					(blank)				
(marked missing)	0.78%	5.23%	1.65%	0.60%	(marked missing)					(marked missing)				
All	3.48%	5.52%	3.16%	3.86%	All	0.88x	1.38x	0.92x	1.00x	All				
FICO x CHANNEL: RANR per booked dollar (more is better)					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	10.18%	24.47%	10.36%	11.00%	under 654	1.43x	1.43x	1.45x	2.11x	under 654	0.59x	2.38x	2.11x	
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x	654 to under 686	1.13x	0.97x	0.92x	
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x	686 to under 712	1.16x	1.08x	1.08x	
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x	712 to under 746	1.15x	0.79x	1.07x	
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x	746 and over	1.20x	0.88x	0.90x	
(blank)					(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)					(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x	All				
FICO x CHANNEL: Outcome, share of booked dollars					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	10.18%	24.47%	10.36%	11.00%	under 654	1.43x	1.43x	1.45x	2.11x	under 654	0.59x	2.38x	2.11x	
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x	654 to under 686	1.13x	0.97x	0.92x	
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x	686 to under 712	1.16x	1.08x	1.08x	
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x	712 to under 746	1.15x	0.79x	1.07x	
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x	746 and over	1.20x	0.88x	0.90x	
(blank)					(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)					(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x	All				

Step 15: Materiality

Materiality

What each materiality level would keep: how many pockets, and how much of the grid's excess they hold. The level itself is set on Control.

FICO x CHANNEL: Outcome, share of loans

In use: 5.7 loans

Share of the book's loans with the outcome	Excess at or over (loans)	Pockets kept	Share of the grid's excess
0.5%	2.9	7	100%
1.0%	5.7	4	92%
2.0%	11.4	3	86%
5.0%	28.5	1	63%
10.0%	57.0	1	63%

FICO x CHANNEL: Outcome, share of booked dollars

In use: 178,222 ORIG_BAL dollars

Share of the book's ORIG_BAL on loans with the	Excess at or over (ORIG_BAL dollars)	Pockets kept	Share of the grid's excess
0.5%	89,111	5	97%
1.0%	178,222	4	94%
2.0%	356,445	3	88%
5.0%	891,112	1	65%
10.0%	1,782,225	1	65%

FICO x CHANNEL: GCO per booked dollar

In use: 99,554 GCO_AMT dollars

Share of the book's total GCO_AMT	Excess at or over (GCO_AMT dollars)	Pockets kept	Share of the grid's excess
0.5%	49,777	6	99%
1.0%	99,554	4	95%
2.0%	199,107	3	87%

Right: for FICO x CHANNEL: Outcome, share of loans, the level in use is 5.7 loans (1% of the book's bad loans).

Step 16: Check

Check records what the run did, including the worked-out numbers (ringed).

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	654; 686; 712; 746 (5 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.34x gap: Outcome, share of loans	about 971
Loans needed for a 1.34x gap: Outcome, share of booked dollars	about 1,219
Loans needed for a 1.34x gap: GCO per booked dollar	about 1,305
Loans needed for a 1.34x gap: RANR per booked dollar	about 228
Materiality line: Outcome, share of loans	5.7 loans
Materiality line: Outcome, share of booked dollars	178,222 ORIG_BAL dollars
Materiality line: GCO per booked dollar	99,554 GCO_AMT dollars
Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars
Worked out from this book	fewest loans 71 (enough to expect 5 with the outcome at the book's rate of 7.13%); worse at 1.34x; better at 0.75x (the outcome gap luck alone can make in a pocket of typical size)
The allowance for many tests covers	each grid and measure on its own, one comparison at a time (the firm's call, 25 Sep 2026)
Revenue counts as more or less	when it's past what luck alone can move that pocket (its own test, at 95% sure)
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	71
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse than its comparison a pocket must be	1.34
How much better than its comparison a pocket must be	0.75
How far revenue must move before it counts as more or less	What luck alone can move it
How sure a difference must be before it counts	95% sure
Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars

Worked out from this book	fewest loans 71 (enough to expect 5 with the outcome at the book's rate of 7.13%); worse at 1.34x; better at 0.75x (the outcome gap luck alone can make in a pocket of typical size)
The allowance for many tests covers	each grid and measure on its own, one comparison at a time (the firm's call, 25 Sep 2026)
Revenue counts as more or less	when it's past what luck alone can move that pocket (its own test, at 95% sure)
How old a loan must be to count	Every loan

Right:

- "Worked out from this book: fewest loans 71 (enough to expect 5 with the outcome at the book's rate of 7.13%); worse at 1.34x; better at 0.75x".
- "Revenue counts as more or less: when it's past what luck alone can move that pocket (its own test, at 95% sure)".
- "132 of 132 agree".

The record beside the workbook (... - what ran.yaml) opens with "# revenue_line: each pocket's own luck range (its own test)".

Step 17: Log, and Start here

Log

Every Run and every refusal, newest first.

2026-09-25 20:36 Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

2026-09-25 20:36 Couldn't run. Fix these, save, close, and press Run again:
Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.

Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C21: "How far revenue must move before it counts as more or less" needs an answer. Pick one, or enter your own in column D.
Control!C23: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.

Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Start here now shows 0 calls, and the last run:

Origination Cube

Last run on 8,000 loans from Consumer book Q3.csv.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.
Where things stand		
	Calls still to make on Control	0
	Things to look at first on Columns	0
	Odd values found	2
	Last run	2026-09-25 20:36: Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Step 18: Split every pocket by a number (revolving debt)

On **Columns**, set **Split pockets by it?** to **Yes** on REV_DEBT. Save, close, and Run.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan name contains 'bad'; 0 and 1 (7.1% are 1)	0%	7655.75, 27999.47, 39684.43 0, 2, 1
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes		Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)				
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes		Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.

Origination Cube

Extract: -5da1-bd67-4db589c3cc31/scratchpad/w7/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 581 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Nothing is worse for RANR per booked dollar at these settings.
Worked out from this book: fewest loans 71; worse at 1.34x; better at 0.75x.
Split by REV_DEBT: each pocket halved at its own median. See the Split and Three-way tabs.
REV_DEBT isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx: start with where it bleeds.

Right: "581 tie-out checks agree" and "Split by REV_DEBT: each pocket halved at its own median."

Split opens with **How this tab works**. The Luck alone row says which figures carry the allowance for many tests (the heat maps) and which don't (the summary, one pooled test per grid and measure). What it assumes explains the correlation once.

Split

Every pocket split in two at its own median REV_DEBT. How the tab works comes first; then each grid: what it holds fixed, a summary, and the pockets as heat maps.

How this tab works

What it does

Inside each pocket (one band, one segment) the loans are sorted by REV_DEBT and cut at that pocket's own median. The high half is compared with the low half, so the band and segment are the same on both sides. REV_DEBT isn't cut into bands of its own while it splits.

High half vs low

The high half's rate divided by the low half's. 2.00x means the high half goes bad, or loses, twice as often. For RANR, above 1.00x means the high half earns more.

Luck alone

How often a gap this big turns up by chance when there's no real difference. The test weighs the gap against how much each half's rate wobbles at its size. The pocket figures (the heat maps) are after the allowance for many tests (Benjamin-Hochberg), across the pockets of one grid and one measure; a pocket whose gap could be luck shows its multiple in brackets, unshaded. The summary's figure is one pooled test per grid and measure, with no allowance. Blank: a half has fewer loans or losses than the minimums on Control (71 loans, 10 losses).

Pooled across pockets

The high halves' actual total against what it would be at their low halves' rates, added over every pocket, with its range at 95% sure. For the yes/no outcome only, the odds are pooled too (Mantel-Haenszel: a standard way to combine pockets without mixing their loans), and Cochran's Q checks whether the gap is about the same size in every pocket.

What it assumes

A grid holds fixed only its band and segment. Anything REV_DEBT moves with that the grid doesn't hold fixed can show up here as a REV_DEBT effect, so each grid gives the correlation, and grids that hold fixed what REV_DEBT moves with most come first. A correlation runs from -1 to 1: 0 means unrelated, and the further from 0, the more of a gap may belong to the other column. Inside a band the score still varies a little, so a little can remain even there. Loans are treated as independent of each other.

FICO x CHANNEL

This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed (REV_DEBT's correlation with it: -0.02).

Measure	Pockets tested	High half worse in	High vs low, pooled	Range (95%)	Luck alone	As odds	Same size in every pocket?
Outcome, share of loans	15	14 of 15	1.95x	1.72x to 2.19x	under 0.01%	2.10x	yes
Outcome, share of booked dollars	15	15 of 15	2.07x	1.80x to 2.34x	under 0.01%		yes/no outcome only
GCO per booked dollar	15	14 of 15	2.06x	1.78x to 2.33x	under 0.01%		yes/no outcome only
RANR per booked dollar	15	5 of 15	1.05x	0.98x to 1.13x	17.7%		yes/no outcome only

Each grid then gets one line on what it holds fixed, a summary and heat maps. **A multiple in brackets, unshaded, could be luck.** For FICO x CHANNEL the high-debt half goes bad **1.95 times** as often as the low half (1.72x to 2.19x), worse in 14 of 15 pockets. The planted effect is 1.8x.

FICO x CHANNEL

This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed (REV_DEBT's correlation with it: -0.02).

Measure	Pockets tested	High half worse in	High vs low, pooled	Range (95%)	Luck alone	As odds	Same size in every pocket?
Outcome, share of loans	15	14 of 15	1.95x	1.72x to 2.19x	under 0.01%	2.10x	yes
Outcome, share of booked dollars	15	15 of 15	2.07x	1.80x to 2.34x	under 0.01%		yes/no outcome only
GCO per booked dollar	15	14 of 15	2.06x	1.78x to 2.33x	under 0.01%		yes/no outcome only
RANR per booked dollar	15	5 of 15	1.05x	0.98x to 1.13x	17.7%		yes/no outcome only

Outcome, share of loans	Branch	Broker	Online
under 654	2.39x	2.67x (1.63x)	2.26x
654 to under 686	(1.27x)		
686 to under 712	(2.12x)	(1.90x)	(1.56x)
712 to under 746	(2.14x)	(0.78x)	(1.44x)
746 and over	(1.30x)	(2.33x)	(1.12x)
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.92%	under 0.01%	14.6%
654 to under 686	39.2%	0.54%	4.4%
686 to under 712	14.6%	17.5%	39.2%
712 to under 746	15.2%	66.2%	48.1%
746 and over	60.3%	14.6%	80.9%
(blank)			
(marked missing)			

Outcome, share of booked dollars	Branch	Broker	Online
under 654	2.59x	2.11x (2.02x)	2.83x
654 to under 686	(1.58x)	3.82x	2.83x
686 to under 712	(2.39x)	(2.69x)	(1.30x)
712 to under 746	(2.68x)	(1.12x)	(1.63x)
746 and over	(1.17x)	(2.82x)	(1.01x)
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.46%	0.03%	5.1%
654 to under 686	23.9%	0.26%	1.6%
686 to under 712	8.7%	7.9%	70.6%
712 to under 746	8.7%	90.1%	41.1%
746 and over	84.5%	8.7%	98.8%
(blank)			
(marked missing)			

GCO per booked dollar	Branch	Broker	Online
under 654	3.02x	2.08x (2.04x)	3.04x
654 to under 686	(1.40x)		
686 to under 712	(2.27x)	(3.06x)	(1.74x)
712 to under 746	(2.75x)	(1.02x)	(1.35x)
746 and over	(1.30x)	(3.09x)	(0.89x)
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.41%	0.12%	6.3%
654 to under 686	44.7%	0.73%	1.4%
686 to under 712	12.1%	6.3%	36.6%
712 to under 746	8.2%	97.1%	66.6%
746 and over	66.6%	8.0%	87.5%
(blank)			
(marked missing)			

RANR per booked dollar	Branch	Broker	Online
under 654	(0.96x)	(1.19x)	(0.97x)
654 to under 686	(0.97x)	(1.00x)	(1.21x)

Luck alone	Branch	Broker	Online
under 654	99.3%	95.5%	99.3%
654 to under 686	99.3%	99.4%	95.5%

Grids that don't hold FICO fixed come after a dark red heading. **Don't quote the numbers under it.** On a book with no debt effect they still read 1.62x, luck alone under 0.01% (defect 8).

Grids that don't hold FICO fixed, REV_DEBT moves with FICO (correlation -0.53), so part of every gap below may be FICO, not REV_DEBT.

ORIG_BAL x CHANNEL

This grid holds ORIG_BAL fixed (REV_DEBT's correlation with it: -0.02). It doesn't hold FICO fixed (REV_DEBT's correlation with it: -0.53).

Measure	Pockets tested	High half worse in	High vs low, pooled	Range (95%)	Luck alone	As odds	Same size in every pocket?
Outcome, share of loans	15	15 of 15	2.55x	2.26x to 2.83x	under 0.01%	2.73x	yes
Outcome, share of booked dollars	15	15 of 15	2.59x	2.27x to 2.91x	under 0.01%		yes/no outcome only
GCO per booked dollar	15	15 of 15	2.61x	2.28x to 2.95x	under 0.01%		yes/no outcome only
RANR per booked dollar	15	7 of 15	1.02x	0.94x to 1.09x	64.9%		yes/no outcome only

Three-way ranks every band / segment / half pocket. The last column, **Holds FICO fixed?**, reads yes or no: part of this may be FICO, and the no rows come last in each measure.

Three-way

Pockets are the REV_DEBT, each tested the other, testing more than their share of REV_DEBT's share of loans. The table compares each pocket with the rest of its band. Asset class, Broker, Online, Word, Out could be luck, Branch, Broker, Online, but two line loans or loans to book, or worth a book by hand. "Luck alone" is how often a gap this big turns up with no real difference behind it, after the difference for testing more pockets within each gap than FICO correlation. 0.01% Bank shows grid doesn't hold FICO fixed (line and column gaps not same after the test, and part of their gap may be FICO).

Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Up rest of book	Luck alone	Up rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show	Holds FICO fixed?
Outcome, share of loans	FICO	under 654	CHANNEL / REV_DEBT	Broker / REV_DEBT high half	205	25.55%	7.13%	7.44 loans	yes	yes	2.73x	under 0.01%	2.26x	under 0.01%	2.55x	1.63x or more yes	1.63x or more yes
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	4 / REV_DEBT high half	196	25.55%	7.13%	40.6 loans	yes	yes	4.17x	under 0.01%	2.58x	0.94% worse	2.58x	1.63x or more yes	1.63x or more yes
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	3 / REV_DEBT high half	210	26.48%	7.13%	28.0 loans	yes	yes	3.03x	under 0.01%	1.44x	14.2% worse, but could be luck	1.44x	1.77x or more yes	1.77x or more yes
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	3 / REV_DEBT high half	175	26.00%	7.13%	22.5 loans	yes	yes	2.82x	under 0.01%	1.38x	21.7% worse, but could be luck	1.38x	1.68x or more yes	1.68x or more yes
Outcome, share of loans	FICO	under 654	CHANNEL / REV_DEBT	Branch / REV_DEBT high half	251	14.34%	7.13%	18.1 loans	yes	yes	2.08x	0.27%	0.94x	82.0% in line	0.94x	1.70x or more yes	1.70x or more yes
Outcome, share of loans	FICO	under 654	CHANNEL / REV_DEBT	Broker / REV_DEBT high half	262	13.38%	7.13%	16.3 loans	yes	yes	1.83x	0.58%	0.87x	62.1% in line	0.87x	1.68x or more yes	1.68x or more yes
Outcome, share of loans	FICO	under 654	CHANNEL / REV_DEBT	Online / REV_DEBT high half	265	12.83%	7.13%	15.1 loans	yes	yes	1.85x	0.99%	0.83x	46.1% in line	0.83x	1.68x or more yes	1.68x or more yes
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	2 / REV_DEBT high half	197	14.12%	7.13%	15.0 loans	yes	yes	2.12x	0.61%	0.98x	98.6% in line	0.98x	1.70x or more yes	1.70x or more yes
Outcome, share of loans	FICO	654 to under 686	ASSET_CLASS / REV_DEBT	4 / REV_DEBT high half	205	14.15%	7.13%	14.4 loans	yes	yes	2.04x	0.80%	1.93x	5.1% worse, but could be luck	1.93x	1.78x or more yes	1.78x or more yes

below the line	1.17x	71.2%	1.01x	99.8% in line	1.78x or more yes
below the line	1.11x	70.4%	0.48x	0.01% better	1.67x or more yes
below the line	1.09x	81.4%	1.71x	29.4% worse, but could be luck	1.80x or more yes
below the line	1.06x	83.5%	0.90x	76.0% in line	1.69x or more yes
below the line	1.08x	81.4%	0.47x	0.10% better	1.80x or more yes
below the line	1.04x	86.8%	1.66x	31.9% worse, but could be luck	1.69x or more yes
yes	2.53x	under 0.01%	2.38x	0.04% worse	1.67x or more no: part of this may be FICO
yes	2.46x	0.07%	2.62x	0.08% worse	1.77x or more no: part of this may be FICO
yes	1.97x	0.72%	2.54x	0.11% worse	1.69x or more no: part of this may be FICO
yes	2.18x	0.50%	2.87x	0.09% worse	1.78x or more no: part of this may be FICO
yes	1.91x	0.82%	1.96x	0.87% worse	1.68x or more no: part of this may be FICO
yes	1.89x	0.99%	3.02x	0.04% worse	1.68x or more no: part of this may be FICO
yes	1.83x	1.1%	2.43x	0.13% worse	1.66x or more no: part of this may be FICO
yes	2.10x	0.74%	1.81x	3.3% worse	1.78x or more no: part of this may be FICO

Step 19: Split every pocket by a category (asset class)

Clear REV_DEBT's split and set **ASSET_CLASS** instead. Save, close, and Run.

Origination Cube

Extract: -5da1-bd67-4db589c3cc31/scratchpad/w7/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 382 tie-out checks agree.
 Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
 Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
 Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
 Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
 Worked out from this book: fewest loans 71; worse at 1.31x; better at 0.76x.
 Split by ASSET_CLASS: one layer per value. See the Three-way tab. ASSET_CLASS isn't cut on its own while it splits.
 Open Consumer book Q3 - Origination Cube.xlsx: start with where it bleeds.

Right: "Split by ASSET_CLASS: one layer per value." On **Three-way**, FICO under 654 / Broker / asset class 4 (131 loans, 29% bad) is flagged **worse**, 2.11x the rest of its band.

Three-way

Pockets table by ASSET_CLASS, each tested like any other pocket, losing more than their share (RANR: earning less, target fine). The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. Blue: material. But too few loans or issues to test, so worth a look by hand. "Luck alone" is how often a gap this big turns up with no real difference behind it, after the allowance for testing many pockets within each grid (see Check).

Measure	Band column		Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 4	131	29.01%	7.13%	28.7 loans		yes	4.2% under 0.01%	2.11x	0.45% worse			1.99x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 4	124	27.42%	7.13%	25.2 loans		yes	4.01x under 0.01%	2.02x	1.9% worse			2.02x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 3	111	29.73%	7.13%	25.1 loans		yes	4.37x under 0.01%	2.14x	0.55% worse			2.07x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 1	145	22.07%	7.13%	21.7 loans		yes	3.22x under 0.01%	1.54x	16.1% worse, but could be luck			1.94x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 1	131	23.66%	7.13%	21.7 loans		yes	3.45x under 0.01%	1.71x	7.4% worse, but could be luck			1.99x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 3	120	24.17%	7.13%	26.4 loans		yes	3.52x under 0.01%	1.74x	7.4% worse, but could be luck			2.04x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 2	136	18.38%	7.13%	15.3 loans		yes	2.65x	0.28%	1.25x	65.6% in line		1.97x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 4	146	15.75%	7.13%	17.6 loans		yes	2.76x	1.6%	1.06x	68.4% in line		1.94x or more

Step 20: Show a median or average per pocket

On **Columns**, set **Show per pocket** to average on ORIG_BAL and median on REV_DEBT. Save, close, and Run. Each grid on **Grids** gets a table of it, marked "not tested".

FICO x CHANNEL: average ORIG_BAL per pocket
Beside the rates, for reading them: not tested, and an average doesn't add up across pockets.

Average	Branch	Broker	Online	All
under 654	32,549	32,583	32,707	32,614
654 to under 686	32,159	32,621	31,389	32,066
686 to under 712	32,699	32,967	32,345	32,661
712 to under 746	30,751	31,887	32,569	31,747
746 and over	31,757	31,970	32,765	32,162
(blank)				
(marked missing)	34,726	30,898	33,613	33,052
All	32,022	32,370	32,390	32,263

FICO x CHANNEL: median REV_DEBT per pocket
Beside the rates, for reading them: not tested, and a median doesn't add up across pockets.

Median	Branch	Broker	Online	All
under 654	17,209	16,277	17,083	16,904
654 to under 686	13,556	13,291	13,669	13,535
686 to under 712	12,292	12,169	12,233	12,219
712 to under 746	9,952	10,431	10,659	10,272
746 and over	7,430	6,911	7,234	7,205
(blank)				
(marked missing)	9,964	10,431	10,267	10,326
All	11,443	11,394	11,328	11,393

FICO x ASSET_CLASS: Outcome, share of loans
Loans where BAD_FLAG is 1, as a share of all loans

Rate	1	2	3	4	All
under 654	13.57%	11.20%	15.14%	20.41%	15.05%
654 to under 686	7.30%	5.92%	8.22%	11.22%	8.25%
686 to under 712	4.57%	3.04%	5.41%	6.63%	4.93%
712 to under 746	3.05%	3.48%	2.55%	5.62%	3.72%
746 and over	3.62%	4.01%	4.16%	4.02%	3.95%
(blank)		0.00%			0.00%

Vs the book	1	2	3	4	All
under 654	1.90x	1.57x	2.13x	2.86x	2.11x
654 to under 686	1.03x	0.83x	1.15x	1.57x	1.16x
686 to under 712	0.64x	0.43x	0.76x	0.93x	0.69x
712 to under 746	0.43x	0.49x	0.36x	0.79x	0.52x
746 and over	0.51x	0.56x	0.58x	0.56x	0.55x
(blank)					

Vs the rest of	1	2	3	4	All
under 654					
654 to under 686					
686 to under 712					
712 to under 746					
746 and over					
(blank)					

Step 21: Your own band edges, a band width, and blue rows

Edges: type 620; 680; 740 in FICO's **Band edges** cell, with semicolons. Save, close, and Run.

Origination Cube

Extract:

1. Set up from this extract 2. Run the cube

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.

Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.

Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.

Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.

Worked out from this book: fewest loans 71; worse at 1.33x; better at 0.75x.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: the worst pocket is **FICO under 620 / Broker:** 176 loans, GCO 5.35x the rest of its band. On Losses vs revenue it reads **Losing more, earning the same**, shaded red (RANR 1.17x, which its own test calls luck).

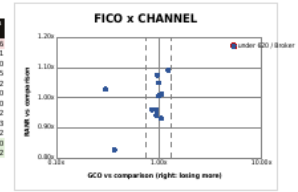
Losses vs revenue

Each pocket's GCO and RANR against the rest of its bands; the bands are against a fair, new counter on more or 1 bin or above and less or 1 bin or below. Revenue counts as more or less only when it's past what's above can move that pocket (its own loss, at 95% sure, both are set on Control. A side past to see that its own loss could be lost is marked, and the row isn't shaded. RANR already includes credit issues, so a pocket earning more is earning more after its losses, and nothing is vetted.

FICO x CHANNEL

Pockets per box: Losing more, earning the same: 1; About the same as both: 9; Losing less, earning the same: 2

Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
under 620	Broker	176	5.35x worse		1.17x about the same		Losing more, earning the same	-4,213,274	-2,561
620 to under 680	Broker	779	1.25x in line		0.93x about the same		About the same on both	18,936	3,810
680 to under 740	Branch	1068	1.07x in line		1.01x about the same		About the same on both	55,552	-1,255
740 to under 760	Online	1137	1.00x in line		1.05x about the same		About the same on both	-884	-5,462
760 to under 800	Branch	727	0.99x in line		1.05x about the same		About the same on both	-6,984	-300
800 to under 820	Online	616	0.95x in line		0.96x about the same		About the same on both	-20,885	2,830
820 to under 860	Online	727	0.96x in line		1.07x about the same		About the same on both	-48,888	-5,242
860 to under 900	Broker	1029	0.94x in line		0.94x about the same		About the same on both	-54,401	6,473
900 to under 940	Online	602	0.85x in line		0.96x about the same		About the same on both	-40,826	2,842
940 to under 980	Branch	184	0.75x better		0.83x about the same		Losing less, earning the same	895,247	3,560
980 to under 1000	Online	207	0.30x better		1.03x about the same		Losing less, earning the same	-880,706	-602



FICO x CHANNEL

Pockets per box: Losing more, earning the same: 1; About the same on both: 9; Losing less, earning the same: 2

Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
under 620	Broker	176	5.35x worse		1.17x about the same		Losing more, earning the same	-4,213,274	-2,561

A width: type every 20 instead.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes	every 20		160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)	Yes			1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'bad'; 0 and 1 (7.1% are 1)	0%	0, 2, 1
GCO_AMT	GCO dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes			suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes				numbers with 7,889 values	0%	1568.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)				
FICO	FICO score	Yes	every 20		160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL	Booked amount	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
BAD_FLAG	Outcome (yes/no)	Yes			1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
GCO_AMT	GCO dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
RANR_AMT	RANR dollars				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
ASSET_CLASS	Category	Yes			suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes			

Origination Cube

Extract: -5da1-bd67-4db589c3cc31/scratchpad/w7/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: REV_DEBT 17,187 and over / CHANNEL Broker.

Worst for Outcome, share of booked dollars: REV_DEBT 17,187 and over / CHANNEL Broker.

Worst for GCO per booked dollar: REV_DEBT 17,187 and over / CHANNEL Broker.

Nothing is worse for RANR per booked dollar at these settings.

13 pockets are material but too small to test: shaded blue on Where it bleeds, to look at by hand.

Worked out from this book: fewest loans 71; worse at 1.57x; better at 0.75x.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: the window names REV_DEBT 17,187 and over / Broker as the worst pocket, by excess, and says "13 pockets are material but too small to test: shaded blue on Where it bleeds". The 13 are rows, one per measure: 5 pockets.

On **Where it bleeds**, FICO 600 to under 620 / Broker (99 loans, 50.5% bad) is flagged **worse**. 580 to under 600 / Broker (50 loans, 29 bad) is under the 71-loan floor and is **blue**: look at it by hand.

Where it bleeds

Pockets losing more lose their share (RANR: earning less), largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. Blue: material, but too few loans or losses to test, so worth a look by hand. "Luck alone" is how often a gap this big turns up with no real difference behind it, after the allowance for testing many pockets within each grid (see Checks).

Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Broker	507	22.29%	7.13%	76.9 loans	yes		3.65x under 0.01%	2.01x under 0.01% worse				1.48x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET CLASS	4	424	18.63%	7.13%	48.8 loans	yes		2.67x under 0.01%	1.03x	8.2% worse, but could be luck			1.53x or more
Outcome, share of loans	FICO	600 to under 620	CHANNEL	Broker	99	50.51%	7.13%	42.9 loans	yes		7.67x under 0.01%	4.25x under 0.01% worse				2.16x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET CLASS	1	411	14.36%	7.13%	29.7 loans	yes		2.13x under 0.01%	0.98x	95.1% in line			1.53x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET CLASS	3	380	14.21%	7.13%	26.9 loans	yes		2.10x	0.01%	0.96x	93.2% in line		1.56x or more
Outcome, share of loans	FICO	580 to under 600	CHANNEL	Broker	50	58.00%	7.13%	25.4 loans	yes		8.52x		5.63x	too few loans to test		2.71x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	CHANNEL	Broker	537	11.73%	7.13%	24.7 loans	yes		1.73x	0.58%	1.58x	5.6% worse, but could be luck		1.46x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Branch	525	11.62%	7.13%	23.6 loans	yes		1.71x	0.14%	0.72x	6.3% better, but could be luck		1.47x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Online	568	10.56%	7.13%	19.3 loans	yes		1.54x	0.75%	0.63x	0.22% better		1.45x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET CLASS	4	424	18.63%	7.13%	48.8 loans	yes		2.87x under 0.01%	1.41x	8.7% worse, but could be luck			1.53x or more
Outcome, share of loans	FICO	600 to under 620	CHANNEL	Broker	99	50.51%	7.13%	42.9 loans	yes		7.67x under 0.01%	4.25x under 0.01% worse				2.16x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET CLASS	1	411	14.36%	7.13%	29.7 loans	yes		2.13x under 0.01%	0.98x	95.1% in line			1.53x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET CLASS	3	380	14.21%	7.13%	26.9 loans	yes		2.10x	0.01%	0.96x	93.2% in line		1.56x or more
Outcome, share of loans	FICO	580 to under 600	CHANNEL	Broker	50	58.00%	7.13%	25.4 loans	yes		8.52x		5.63x	too few loans to test		2.71x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	CHANNEL	Broker	537	11.73%	7.13%	24.7 loans	yes		1.73x	0.58%	1.58x	5.6% worse, but could be luck		1.46x or more

On Losses vs revenue, 600 to under 620 / Broker reads Losing more, earning the same:

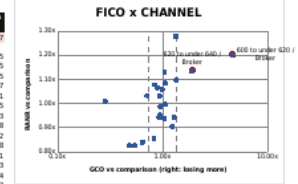
Losses vs revenue

Each pocket's loss and RANR against the rest of its loans; the loans are against a line, your counts are more or less or above and less or below. Revenue counts as more or less only when the gain about loss above can move that pocket (its own loss, at 95% sure, both are set on Control, a side panel to see that its own test case could be luck is marked, and the rest isn't shaded. RANR already includes credit losses, so a pocket earning more is earning more after its losses, and nothing is wrong.

FICO x CHANNEL

Pockets per box: Losing more, earning the same: 1; About the same as both: 16; Losing less, earning the same: 2; Not tested: too few loans or losses: 7; Marked could be luck, not counted above: 4

Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
600 to under 620	Broker	99	4.63x worse		1.20x about the same		Losing more, earning the same	755,503	-3,687
620 to under 640	Broker	174	1.92x worse, but could be luck		1.14x about the same		Losing more, earning the same (loss gap could be luck)	191,863	-2,645
640 to under 660	Online	249	1.34x in line		1.28x about the same		About the same on both	129,626	-6,255
660 to under 680	Broker	342	1.24x in line		0.90x about the same		About the same on both	115,918	3,375
680 to under 700	Branch	234	1.25x in line		0.94x about the same		About the same on both	110,477	1,387
700 to under 720	Branch	367	1.35x in line		1.09x about the same		About the same on both	80,385	-3,401
720 to under 740	Online	395	1.06x in line		0.99x about the same		About the same on both	22,210	195
740 to under 760	Online	334	1.05x in line		1.08x about the same		About the same on both	12,981	-2,913
760 to under 780	Online	226	1.05x in line		1.13x about the same		About the same on both	8,222	-3,198
780 to under 800	Broker	250	1.04x in line		0.94x about the same		About the same on both	1,161	1,692
800 to under 820	Broker	361	1.00x in line		1.06x about the same		About the same on both	-1,709	-2,008
820 to under 840	Branch	325	0.95x in line		0.99x about the same		About the same on both	-11,693	471
840 to under 860	Branch	238	0.92x in line		0.94x about the same		About the same on both	-15,427	1,553
860 to under 880	Branch	376	0.95x in line		0.95x about the same		About the same on both	-20,112	1,884
880 to under 900	Branch	317	0.94x in line		1.03x about the same		About the same on both	-31,423	-932
900 to under 920	Online	408	0.89x in line		1.07x about the same		About the same on both	-33,338	-2,678
920 to under 940	Branch	378	0.83x in line		0.85x about the same		About the same on both	-51,405	6,196
940 to under 960	Online	309	0.84x in line		1.08x about the same		About the same on both	-78,059	-2,254
960 to under 980	Branch	176	0.71x better, but could be luck		1.03x about the same		Losing less, earning the same (loss gap could be luck)	-87,066	-655



FICO x CHANNEL

Pockets per box: Losing more, earning the same: 1; About the same on both: 16; Losing less, earning the same: 2; Not tested: too few loans or losses: 7; Marked could be luck, not counted above: 4

Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR reading	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
600 to under 620	Broker	99	4.63x worse		1.20x about the same		Losing more, earning the same	755,503	-3,687

A high floor. Type your own fewest loans, 450, on Control. Pockets under it aren't tested, and the material ones turn blue. The window says "58 pockets are material but too small to test"; that is 58 rows over four measures, 23 pockets.

Origination Cube

Extract: 7-4db589c3cc31/scratchpad/w7/wrong/min450/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.

Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.

Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.

Worst for RANR per booked dollar: REV DEBT 6,559 to under 9,837 / CHANNEL Broker.

58 pockets are material but too small to test: shaded blue on Where it bleeds, to look at by hand.

Worked out from this book: worse at 1.31x; better at 0.76x.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Where it bleeds

Pockets losing more than their share (RANK: earning less), largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. Blue: material, but too few loans or losses to test, so worth a look by hand. "Luck alone" is how often a gap this big turns up with no real difference behind it, after the allowance for testing many pockets within each grid (See Checks).

Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	90.7 loans	yes		4.14x	under 0.01%	2.38x	under 0.01% worse		1.47x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Broker	507	22.29%	7.13%	76.9 loans	yes		3.65x	under 0.01%	2.01x	under 0.01% worse		1.48x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	4	392	20.41%	7.13%	52.1 loans	yes		3.17x		1.54x		too few loans to test	1.55x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	4	424	18.63%	7.13%	48.8 loans	yes		2.87x		1.41x		too few loans to test	1.53x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	1	411	14.36%	7.13%	29.7 loans	yes		2.13x		0.98x		too few loans to test	1.53x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	3	350	15.14%	7.13%	28.1 loans	yes		2.24x		1.01x		too few loans to test	1.58x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	1	420	13.57%	7.13%	27.1 loans	yes		2.01x		0.87x		too few loans to test	1.53x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	3	380	14.21%	7.13%	26.9 loans	yes		2.10x		0.96x		too few loans to test	1.56x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	CHANNEL	Broker	537	11.73%	7.13%	24.7 loans	yes		1.73x	0.58%	1.58x		5.6% worse, but could be luck	1.46x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Branch	525	11.62%	7.13%	23.6 loans	yes		1.71x	0.14%	0.72x		6.3% better, but could be luck	1.47x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL	Online	568	10.56%	7.13%	19.5 loans	yes		1.54x	0.75%	0.63x		0.22% better	1.45x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET_CLASS	4	414	11.59%	7.13%	18.5 loans	yes		1.68x		1.79x		too few loans to test	1.53x or more
Outcome, share of loans	FICO	under 654	CHANNEL	Online	531	10.36%	7.13%	17.2 loans	yes		1.50x	1.8%	0.59x		0.04% better	1.47x or more
Outcome, share of loans	FICO	654 to under 686	ASSET_CLASS	4	410	11.22%	7.13%	16.8 loans	yes		1.62x		1.56x		too few loans to test	1.54x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	2	393	11.20%	7.13%	16.0 loans	yes		1.62x		0.68x		too few loans to test	1.55x or more
Outcome, share of loans	REV_DEBT	13,051 to under 17,187	ASSET_CLASS	4	411	10.95%	7.13%	15.7 loans	yes		1.58x		1.51x		too few loans to test	1.53x or more
Outcome, share of loans	REV_DEBT	13,051 to under 17,187	CHANNEL	Broker	553	9.95%	7.13%	15.6 loans	yes		1.44x	2.8%	1.37x		22.3% worse, but could be luck	1.46x or more
Outcome, share of loans	FICO	under 654	CHANNEL	Branch	501	10.18%	7.13%	15.3 loans	yes		1.47x	2.9%	0.59x		0.04% better	1.48x or more
Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	2	385	10.91%	7.13%	14.6 loans	yes		1.57x		0.69x		too few loans to test	1.55x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET_CLASS	4	413	10.41%	7.13%	13.6 loans	yes		1.50x		1.25x		too few loans to test	1.53x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	CHANNEL	Broker	531	9.23%	7.13%	11.2 loans	yes		1.32x	20.8%	1.30x		25.5% in line	1.47x or more
Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	ASSET_CLASS	4	412	9.71%	7.13%	10.6 loans	yes		1.39x		1.80x		too few loans to test	1.53x or more

Where it bleeds

Pockets losing more than their share (RANK: earning less), largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. Blue: material, but too few loans or losses to test, so worth a look by hand. "Luck alone" is how often a gap this big turns up with no real difference behind it, after the allowance for testing many pockets within each grid (See Checks).

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Outcome, share of loans	REV_DEBT	17,187 and over	ASSET_CLASS	1	411	14.36%	7.13%	29.7 loans	yes		2.13x		0.98x		too few loans to test	1.53x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	3	350	15.14%	7.13%	28.1 loans	yes		2.24x		1.01x		too few loans to test	1.58x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	1	420	13.57%	7.13%	27.1 loans	yes		2.01x		0.87x		too few loans to test	1.53x or more
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Outcome, share of loans	FICO	654 to under 686	ASSET_CLASS	4	410	11.22%	7.13%	16.8 loans	yes		1.62x		1.56x		too few loans to test	1.54x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS	2	393	11.20%	7.13%	16.0 loans	yes		1.62x		0.68x		too few loans to test	1.55x or more
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Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	ASSET_CLASS	4	412	9.71%	7.13%	10.6 loans	yes		1.39x		1.80x		too few loans to test	1.53x or more

Step 22: Press Set up again

Press **1. Set up from this extract** again.

Origination Cube

Extract: -5da1-bd67-4db589c3cc31/scratchpad/w7/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns.

2 columns to look at first, on the Columns tab.

Everything is answered. Press Run the cube.

Right: "Everything is answered. Press Run the cube." Control, Columns and Odd values answers are kept, and so are the last results. **Last Run used** is kept too (ringed):

Control center					
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.					
Shaded = still needs an answer					
Setting	Choose	Or enter your own	In use	What it means	Last Run used
What the cube looks at	Every loan		Every loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.	Every loan
How old a loan must be to count					
How columns are recognised	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.	
A number column with this many values or fewer is a category					
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.	
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.	
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.	
Is it enough loans to matter	Enough for 5 expected losses (suggested)		Enough for 5 expected losses (suggested)	Worked out from this book: the loans a pocket needs to expect 5 with the outcome at the book's rate, the textbook floor for testing a rate.	71 (worked out from this book)
Fewest loans in a pocket before it is tested	10 losses		10 losses	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.	10 losses
Is it material	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.	1% of the book's total losses
Smallest excess loss worth reporting					
Does it underperform	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.	The rest of its band
What a pocket is judged against					
How much worse than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the gap luck alone can make in a pocket of typical size. A smaller gap can't be told from noise.	1.37 (worked out from this book)
How much better than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	The same as above, the other way: one over the gap luck alone can make in a pocket typical size.	0.73 (worked out from this book)
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the revenue gap luck alone can make in a pocket of typical size. Revenue inside that gap reads as about the same.	each pocket's own luck range
Is it real	95% sure		95% sure	The usual standard. About 1 pocket in 20 with no real difference will still pass.	95% sure
How sure a difference must be before it counts					
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.	80% of the time
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.	Hold down the share of lucky finds (Benjamini-Hochberg)

Step 23: Prune something learned

On **Learned**, set a row's **Keep?** (column A) to **Forget**. Save, close and Run.

Learned						
Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run, and the column waits on Columns for you to confirm it again.						
Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	ASSET_CLASS	Category	2026-09-25	2026-09-25	26
Keep	column	BAD_FLAG	Outcome (yes/no)	2026-09-25	2026-09-25	26
Forget	column	CHANNEL	Category	2026-09-25	2026-09-25	26
Keep	column	FICO	fico; band edges every 20	2026-09-25	2026-09-25	26
Keep	column	GCO_AMT	GCO dollars	2026-09-25	2026-09-25	26
Keep	column	LOAN_NBR	Loan number (key)	2026-09-25	2026-09-25	26
Keep	column	ORIG_BAL	Booked amount	2026-09-25	2026-09-25	26
Keep	column	RANR_AMT	RANR dollars	2026-09-25	2026-09-25	26
Keep	column	REV_DEBT	Amount or number	2026-09-25	2026-09-25	26
Keep	column	FICO	0000-9999 missing	2026-09-25	2026-09-25	26

Right: the window says "Forgot CHANNEL, as marked on Learned. Check it on Columns and set C3 to Yes before the next Run."

Origination Cube

Extract: -5da1-bd67-4db589c3cc31/scratchpad/w7/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: REV_DEBT 17,187 and over / CHANNEL Broker.
Worst for Outcome, share of booked dollars: REV_DEBT 17,187 and over / CHANNEL Broker.
Worst for GCO per booked dollar: REV_DEBT 17,187 and over / CHANNEL Broker.
Nothing is worse for RANR per booked dollar at these settings.
13 pockets are material but too small to test: shaded blue on Where it bleeds, to look at by hand.
Worked out from this book: fewest loans 71; worse at 1.37x; better at 0.73x.
Forgot CHANNEL, as marked on Learned. Check it on Columns and set C3 to Yes before the next Run.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

The next Run refuses until you do:

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!C3: set Checked every column to Yes once you've checked each column's meaning. Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again.

Step 24: The next extract

Browse to the next quarter's extract and press **Set up**.

Origination Cube

Extract: 69-5da1-bd67-4db589c3cc31/scratchpad/w7/q4/Consumer book Q4.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q4 - Origination Cube.xlsx from Consumer book Q4.csv: 6,000 loans, 9 columns.

3 columns to look at first, on the Columns tab.

Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

Right: "3 columns to look at first". Meanings are remembered, and so are band edges from the last Run. FICO comes in as every 20, shaded, with "Band edges remembered from before: every 20." under **Look first**. It is one of the three:

Columns								
For any meaning that's wrong, and set Cut by it to No for anything you don't want in the grid. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.								
Checked every column?								
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes	every 20		Band edges remembered from before: every 20.	Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 694, 713
CHANNEL	Category	Yes				Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Broker, Branch, Online
ORIG_BAL	Booked amount	Yes				Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	29850.78, 29562.62, 58061.68
BAD_FLAG	Outcome (yes/no)				1 value is neither 0 nor 1. They're left out of the outcome rates and counted, if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0%	0, 2, 1
GCO_AMT	GCO dollars				1 value isn't a number among 5,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0%	0.0, #N/A, 6604.12
RANR_AMT	RANR dollars					Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	107.15, 285.79, 299.02
ASSET_CLASS	Category	Yes				Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	2, 4, 1
REV_DEBT	Amount or number	Yes				Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	18812.23, 6699.74, 10834.7
Checked every column?								
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested		
LOAN_NBR	Loan number (key)					Remembered: you confirmed this as loan or application number on 2026-09-25		
FICO	FICO score	Yes	every 20		Band edges remembered from before: every 20.	Remembered: you confirmed this as a FICO score on 2026-09-25		
CHANNEL	Category	Yes				Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25		

Control starts blank again. The judgment calls are asked every time.

Step 25: What the memory holds

The memory is one file. Every Run writes each band column's edges into it, and a blank cell clears them, so the next new workbook gets whatever the last Run of any workbook had. Edges typed on a category (CHANNEL) are no longer remembered, but they are still accepted without a word and ignored.

Origination Cube

Extract:

Set up Consumer book Q1 - Origination Cube.xlsx from Consumer book Q1.csv: 5,000 loans, 9 columns.
3 columns to look at first, on the Columns tab.
Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

Clear anything under Look first you didn't type.

When something goes wrong

What you did	What the window says	What to do
A. Split set on two columns	"Columns!H13 and Columns!H14: only one column can split the pockets. Clear all but one."	Clear one.
B. Split the only category that's cut	"Nothing is left to cut across ... CHANNEL (Columns!H8: it splits the pockets ...); ASSET_CLASS (Columns!D13, Cut by it? No)."	Clear the split, or set ASSET_CLASS back to Yes.
C. average on a category	"Columns!G8: "CHANNEL" isn't a number column, so it has no average to show."	Clear the cell.
		Retype with semicolons.

What you did	What the window says	What to do
D. Edges 620,680,740 turned into one number by Excel	"... read as the single number 620,680,740. Excel dropped the commas. Type them with semicolons"	
D2. An edge outside the column (620; 680; 950)	"FICO runs from 496 to 921, so an edge at 950 would leave a band empty."	Type edges inside that range.
E. Workbook still open in Excel	"... is open in Excel. Close it, then press Run again." Set up says the same.	Close Excel and press again.
F. A column renamed after Set up	"Couldn't run: the extract has no column "CHANNEL" (a segment) ... press Set up again." Set up then says "New columns since the last check: CHNL."	Press Set up, check the new name, set C3 to Yes.
G. Fewer rows after Set up	"Consumer book Q3.csv has changed since Set up ...", then it runs on 6,000.	Check the loan count.
H. Workbook copied to a folder with a different extract of the same name	The same "has changed" line; it runs the new folder's 3,000 loans.	Press Set up in the new folder first.
I. Split set on GCO, the key or the outcome	"Columns!H11: "GCO_AMT" is marked GCO dollars, which can't split the pockets. ..."	Split by something known at booking.
I2. Split by the booked amount	Runs, 571 tie-outs.	Allowed.
J. A dollar amount typed as materiality (100000)	Runs.	Check says it's GCO dollars only.
K. The workbook picked as the extract	Set up: "... is the workbook, not the loan file." Run: "Pick the extract beside it."	Browse to the .csv.
L. 95 typed for "how sure"	"... from 0.5 to 0.999; got 95. For 95%, type 0.95."	Type 0.95.
M. 95, or 0.95, as your own revenue line	95: "... for 15%, type 0.15." 0.95: "The most it takes is 0.9."	Type a share up to 0.9.
N. New columns in a refreshed extract	"New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again."	Check them, set C3 to Yes.
O. Forget on Learned, then Run twice	First Run runs and says what was forgotten; the second refuses on C3.	Check the column, set C3 to Yes.
P. every 1 or every 0.5 on FICO	"Columns!F7: every 1 on "FICO" would make 426 bands (496 to 921). Use a wider band, so there are 50 bands or fewer."	Use a wider width.
Q. every 0, every -5, every twenty	"Columns!F7: "every 0" should read like every 20: the word every, then how wide each band is."	Type a positive number in figures.
R. Edges typed on a category (CHANNEL)	Runs, says nothing. The cell is ignored and no longer remembered.	Clear it.

What you did	What the window says	What to do
S. Your own fewest loans bigger than any pocket (3000), with luck suggested	"No pocket had enough loans or losses to test ..." for every measure, and "Nothing in this book to work these out from, so the usual values were used: worse at 1.25x; better at 0.80x."	Use a smaller number. Check still calls 1.25x the luck gap (defect 6).
T. A small low-default book (5,000 loans, 1.34% bad) with every suggestion	"Worked out from this book: fewest loans 374", then "No pocket had enough loans or losses" for every measure (the largest pocket is 363), "130 pockets are material but too small to test", and the usual values for worse and better.	Read it as nothing was tested. Type your own fewest loans, or accept that this book is too small to test pocket by pocket.
U. Your own fewest loans, 450	Runs; "58 pockets are material but too small to test".	58 rows, 23 pockets. Read the blue rows by hand.

A: two splits

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H13 and Columns!H14: only one column can split the pockets. Clear all but one.

B: the only segment split

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Nothing is left to cut across: mark at least one column as a category (or term) on the Columns tab, with Cut by it set to Yes. Changed from what was suggested: CHANNEL (Columns!H8: it splits the pockets, so it isn't a segment of its own); ASSET_CLASS (Columns!D13, Cut by it? No).

C: average on a category

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!G8: "CHANNEL" isn't a number column, so it has no average to show. Clear the cell, or change what it is.

D and D2: edges

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: the band edges for "FICO" read as the single number 620,680,740. Excel dropped the commas. Type them with semicolons: 620; 680; 740.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: FICO runs from 496 to 921, so an edge at 950 would leave a band empty. Type edges inside that range, with semicolons: 620; 680; 740.

E: workbook open. Simulated by locking the file (chattr +i), since Excel isn't available here.

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Run again.

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Set up again.

F: a column renamed, then Set up

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run: the extract has no column "CHANNEL" (a segment). Its columns are: LOAN_NBR, FICO, CHNL, ORIG_BAL, BAD_FLAG, GCO_AMT, RANR_AMT, ASSET_CLASS, REV_DEBT. If a column was renamed or dropped, press Set up again.

Origination Cube

Extract:

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns.

New columns since the last check: CHNL. Columns!C3 needs a Yes again.

3 columns to look at first, on the Columns tab.

Next: fill in the shaded cells on Columns, save, close, and press Run.

G: fewer rows

Origination Cube

Extract:

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.

Ran on 6,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.

Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.

Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.

Worst for RANR per booked dollar: FICO 714 to under 747 / ASSET_CLASS 2.

Worked out from this book: fewest loans 71; worse at 1.40x; better at 0.71x.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

H: the workbook copied to another folder

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.

Ran on 3,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: FICO under 653 / CHANNEL Broker.

Worst for Outcome, share of booked dollars: FICO under 653 / CHANNEL Broker.

Worst for GCO per booked dollar: FICO under 653 / CHANNEL Broker.

Nothing is worse for RANR per booked dollar at these settings.

Worked out from this book: fewest loans 71; worse at 1.56x; better at 0.64x.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

I and I2: splits

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H11: "GCO_AMT" is marked GCO dollars, which can't split the pockets. Only a score, ratio, amount (the booked amount too) or category can.

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H6: "LOAN_NBR" is marked Loan number (key), which can't split the pockets. Only a score, ratio, amount (the booked amount too) or category can.

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H10: "BAD_FLAG" is marked Outcome (yes/no), which can't split the pockets. Only a score, ratio, amount (the booked amount too) or category can.

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Ran on 8,000 loans from Consumer book Q3.csv; 571 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Worked out from this book: fewest loans 71; worse at 1.34x; better at 0.75x.
Split by ORIG_BAL: each pocket halved at its own median. See the Split and Three-way tabs.
ORIG_BAL isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

J: your own dollar materiality

Origination Cube

Extract: 89c3cc31/scratchpad/w7/wrong/own-materiality/Consumer book Q3.csv

[Browse...](#)

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Worked out from this book: fewest loans 71; worse at 1.34x; better at 0.75x.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

K: the workbook picked, then Run

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is the workbook, not the loan file. Pick the extract it was set up from (Consumer book Q3.csv or Consumer book Q3.xlsx).

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is the workbook, not the loan file. Pick the extract beside it.

L: 95 for "how sure"

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!D23: "How sure a difference must be before it counts" needs a share, such as 0.95 for 95%, from 0.5 to 0.999; got 95. For 95%, type 0.95.

N: new columns

Origination Cube

Extract:

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 11 columns.

New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again.

4 columns to look at first, on the Columns tab.

Next: fill in the shaded cells on Columns, save, close, and press Run.

P: every 1, every 0.5

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: every 1 on "FICO" would make 426 bands (496 to 921). Use a wider band, so there are 50 bands or fewer.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!F7: every 0.5 on "FICO" would make 851 bands (496 to 921). Use a wider band, so there are 50 bands or fewer.

Q: every 0, every -5, every twenty

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: "every 0" should read like every 20: the word every, then how wide each band is.

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: "every -5" should read like every 20: the word every, then how wide each band is.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!F7: "every twenty" should read like every 20: the word every, then how wide each band is.

R: edges on a category

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Worked out from this book: fewest loans 71; worse at 1.34x; better at 0.75x.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

S: a fewest-loans number bigger than any pocket

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
No pocket had enough loans or losses to test Outcome, share of loans (fewest loans: 3,000).
No pocket had enough loans or losses to test Outcome, share of booked dollars (fewest loans: 3,000).
No pocket had enough loans or losses to test GCO per booked dollar (fewest loans: 3,000).
No pocket had enough loans or losses to test RANR per booked dollar (fewest loans: 3,000).
101 pockets are material but too small to test: shaded blue on Where it bleeds, to look at by hand.
Nothing in this book to work these out from, so the usual values were used: worse at 1.25x; better at 0.80x.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

T: a small low-default book

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Ran on 5,000 loans from Consumer book low.csv; 132 tie-out checks agree.
No pocket had enough loans or losses to test Outcome, share of loans (fewest loans: 374).
No pocket had enough loans or losses to test Outcome, share of booked dollars (fewest loans: 374).
No pocket had enough loans or losses to test GCO per booked dollar (fewest loans: 374).
No pocket had enough loans or losses to test RANR per booked dollar (fewest loans: 374).
130 pockets are material but too small to test: shaded blue on Where it bleeds, to look at by hand.
Worked out from this book: fewest loans 374.
Nothing in this book to work these out from, so the usual values were used: worse at 1.25x; better at 0.80x.
Open Consumer book low - Origination Cube.xlsx: start with Where it bleeds.

U: a high floor

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
58 pockets are material but too small to test: shaded blue on Where it bleeds, to look at by hand.
Worked out from this book: worse at 1.31x; better at 0.76x.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

How this was walked

On a Linux test machine:

- **The window:** Origination Cube.pyw's window on a virtual display under Python 3.12, its buttons pressed by driver/drive.py (through driver/win.sh), each state photographed. The file dialog opens behind the window on the virtual display, so it isn't pictured.
- **Excel's part:** done cell by cell with openpyxl (driver/edit.py), standing in for picking from the dropdowns and typing.
- **The workbook:** each state printed through LibreOffice (driver/render.sh) and photographed page by page. driver/shots.py crops and rings every picture.
- **Wrong turns and revenue options:** each in a folder of its own, from a copy of the answered workbook (driver/wrong.sh, and driver/wrong2.sh for the ones that change the extract or the file). They share one scratch memory file with the main route.
- **The build:** frozen at commit a5a8aa2 with git archive.
- **Reading tabs back:** driver/revcheck.py (step 13's table: count lines against rows, shading of marked boxes, RANR reading against the box), driver/bluecheck.py (blue rows against the window's count), driver/threeway.py, driver/luckline.py (what the suggested numbers are made of), driver/noplant.py (the split on a book with no debt effect), driver/lowdefault.py (wrong turn T's book).

- **The next extracts:** `synth.make_rows` with seeds 11 (Q4, 6,000 loans) and 23 (Q1, 5,000 loans).

The order walked: steps 1 to 17, the revenue options on the default bands, 620; 680; 740 and every 20 (each in its own folder), the high floor and the fallback wrong turns, 18 to 22 on the same workbook, 23, 24, 25, then the other wrong turns.